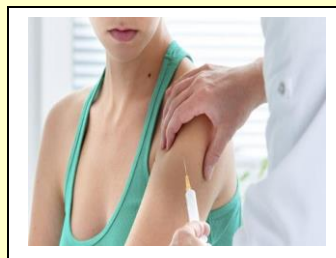




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NATIONAL RABIES CONTROL PROGRAM

***Management of Animal Bites
(Guideline for Medical Officers, Para Medical Staff)***



**Guidelines for use of Intra-dermal inoculation of
Tissue Culture Vaccines**

Epidemic Branch
Commissionerate of Health, Medical Services and Medical Education (H.S.)
Gandhinagar

(Under Guidance of National Center for Disease Control, GoI)

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Introduction

Rabies is an acute viral encephalomyelitis caused by RNA Lyssavirus belonging to Family Rhabdoviridae. The causative agent is found in wild and some domestic animals, and is transmitted to other animals and to humans through close contact with their saliva (i.e. bites, scratches, licks on broken skin and mucous membranes). In urban areas, the disease is mainly transmitted by dogs, being responsible for 95% - 96% of animal bite cases. Man is the dead end of the infection and hence does not play any role in its spread to new hosts.

Rabies has terrified man since antiquity. The fear is by no means unfounded since the disease is invariably fatal and perhaps the most painful and horrible of all communicable diseases in which the sick person is tormented at the same time with thirst and fear of water (hydrophobia). Fortunately, animal bites, if managed appropriately and timely the disease is preventable to a large extent. In this regard the post exposure treatment of animal bite cases is of prime importance.

An expert group meeting on reviewing and finalizing the guidelines for management of animal bite cases was held at National Institute of Communicable Diseases, Delhi. Following this, the national guidelines for management of animal bites were formulated to bring out uniformity in post exposure prophylaxis practices. Until recently the Nervous Tissue Vaccine (NTV) was the mainstay for post exposure prophylaxis. This was being administered free of cost in government hospitals/anti-rabies treatment centers. However, it was associated with serious neuromuscular complications, the incidence of which was reported to vary from 1:5000 to 1:11000 vaccinees.

WHO estimates that more than 55,000 human deaths occur globally due to this disease every year. Rabies continues to be a major public health problem throughout India with the exception of two water logged islands i.e. Andaman & Nicobar and Lakshadweep which are historically free of rabies.

As per WHO recommendation, the production and use of reactogenic vaccine has been stopped since December 2004 in our country. Modern Tissue Culture Vaccines (TCV) are now being used for post exposure prophylaxis. Higher cost of intramuscular administration of TCV is limiting factor for its wider use. To overcome this problem, WHO has recommended use of efficacious, safe and feasible intra-dermal (ID) route of inoculation of TCVs. National authorities after expert consultation have approved the use of id route for administration of TCVs in the country in phased manner.

Post-exposure treatment

Decision to treat

In rabies endemic country like India, where every animal bite is potentially suspected as a rabid animal bite the treatment should be started immediately. Because of long incubation period, which is typical of most cases of human rabies, it is possible to institute prophylactic post exposure treatment. This must be started at the earliest to ensure that the individual will be immunized before the rabies virus reaches the Central Nervous System. However, people who present for treatment even months after a possible rabies exposure should be evaluated and treated as if the event had occurred recently.

To bring out uniformity globally, the classification of animal bite for post-exposure treatment has been based as per WHO recommendations (Table-1).

Table 1: Type of contact, exposure and recommended post-exposure prophylaxis

Category	Type of contact	Type of exposure	Recommended post-exposure prophylaxis
I	Touching or feeding of animals Licks on intact skin	None	None, if reliable case history is available
II	Nibbling of uncovered skin Minor scratches or abrasions without bleeding	Minor	Wound management Antirabies vaccine
III	Single or multiple transdermal bites or scratches, licks on broken skin Contamination of mucous membrane with saliva	Severe	Wound management Rabies immunoglobulin Antirabies vaccine

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Vaccination Status of the biting animal:

Although unvaccinated animals are more likely to transmit rabies, vaccinated animals can also do so if the vaccination of the biting animal was ineffective for any reason. A history of rabies vaccination in an animal is not always a guarantee that the biting animal is not rabid. Animal vaccine failures may occur because of improper administration or poor quality of the vaccine, poor health status of the animal, and the fact that one vaccine dose does not always provide long-lasting protection against infection in dogs. The risk of dog being infected with rabies is greatly reduced when it appears healthy and there is confirmed history of vaccination with minimum of two immunizations with potent rabies vaccine in last two years.

Provoked Vs unprovoked bites:

Whether a dog bite was provoked rather than unprovoked should not be considered a guarantee that the animal is not rabid as it can be difficult to understand what an attacking dog considers provocation for an attack.

Observation of biting animal:

The treatment should be started immediately after the bite. The treatment may be discontinued if animal involved (dog or cat) remains healthy throughout an observation period of 10 days. The observation period is valid for dogs and cats only. The natural history of rabies in mammals other than dogs or cats is not fully understood and therefore the 10-day observation period may not be applicable.

Bite in jungle:

Bite by all wild animals should be treated as category III exposure.

Bite by rodents:

It should be noted that bites by rats, mice, squirrel, hare and rabbits seldom require treatment.

Bat rabies has not been conclusively proved in India and hence exposure does not warrant treatment.

Special Circumstances:

Pregnancy, lactation, infancy, old age and concurrent illness are no contraindications for rabies post-exposure prophylaxis in the event of an exposure. Post-exposure prophylaxis against rabies takes preference over any other consideration since it is a life saving procedure. Moreover, rabies vaccine does not have any adverse effect on

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fetus, mother-to-be and the course of pregnancy. Hence complete post-exposure treatment should be given depending on the category of the exposure.

Post-exposure prophylaxis of immunocompromised patients:

Several studies of patients with HIV/AIDS have reported that those with very low CD4 counts will mount a significantly lower or no detectable neutralizing antibody response to rabies. *In such patients and those in whom the presence of immunological memory is no longer assured as a result of other causes, proper and thorough wound treatment as described above and antisepsis accompanied by local infiltration of a passive immunization product are of utmost importance. Immunocompromised patients with category II exposures should receive rabies immunoglobulin in addition to a full post-exposure vaccination series as listed above. Preferably, if the facilities are available, anti-rabies antibody titration should be performed 10 days after the completion of course of vaccination.*

Human-to-human transmission:

There is no risk to other humans or animals from a patient with rabies. Despite there being tens of thousands of cases each year worldwide, there has never been a documented case of human-to-human transmission, other than the few cases resulting from organ transplant. *Despite the lack of evidence for human-to-human transmission, people who have been exposed closely to the secretions of a patient with rabies will sometimes be offered immunisation purely as a precautionary measure.*

It is re-emphasized that treatment should be started as early as possible after exposure, but it should not be denied to person reporting late for treatment.

Post Exposure Prophylaxis

The post-exposure prophylaxis is a three pronged approach. All three carry equal importance and should be done simultaneously.

- Management of wound
- Passive immunization
- Active immunization

Management of animal bite wound

Wound toilet:

Since the rabies virus enters the human body through a bite or scratch, it is imperative to remove as much saliva, and thereby the virus, from the wound as is possible by an efficient wound toilet that should not involve additional trauma. Since the rabies virus can persist and even multiply at the site of bite for a long time, wound toilet must be performed even if the patient reports late. (Table 2)

This can be done by prompt and gentle thorough *washing with soap or detergent and flushing the wound with running water for 10 minutes. If soap and detergent are not immediately available wash with running water for at least 10 minutes.* Avoid direct touching of wounds with bare hands. Considering the importance of this step the anti rabies clinics should have wound washing facilities.

The application of soil, chilies, oil etc. is unnecessary and damaging. In case soil, chilies, oil etc. have been applied on the wound, enough gentle washing with soap or detergent to remove the extraneous material especially oil should be done followed by flushing with copious amount of water for 10 minutes immediately.

It should be noted that the immediate washing of the wound is a priority. However, the victim should not be deprived of the benefit of wound toilet as long as there is an unhealed wound which can be washed even if the patient reports late. The maximum benefit of the wound washing is obtained when fresh wound is cleaned immediately.

Application of antiseptic:

After thorough washing and drying the wound, any one of the available chemical agents should be applied: Savlon (in appropriate recommended dilution), Dettol (in appropriate recommended dilution), Povidone iodine, alcohol etc.

Local infiltration of rabies immunoglobulin:

In category III bites rabies immunoglobulins should be infiltrated in the depth and around the wound to inactivate the locally present virus. (Section ---)

Suturing of wound should be avoided as far as possible. If unavoidable, minimum loose sutures should be applied after adequate local treatment along with proper infiltration of anti rabies serum.

Cauterization of wound is no longer recommended as it leaves very bad scar, and does not confer any additional advantage over washing the wound with water and soap. Injection **tetanus toxoid** should be given to the un-immunized individual. To

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prevent sepsis in the wound, a suitable course of an **antibiotic** may be recommended.

Table 2: Wound Management

Do's		
Physical	 Wash with running tap water	Mechanical removal of virus from the wound
Chemical	 Washing the wound with soap and water, dry & apply disinfectant	Inactivation of the virus
Biological	 Infiltration of Immuno-globulins in the depth and around the wound in Category III exposures	Neutralization of the virus
Don'ts		
Touch the wound with bare hand Apply irritants like soil, chilies, oil, herbs, chalk, betel leaves etc.		

Passive Immunization by Rabies Immunoglobulin (RIG)

The anti-rabies serum provides passive immunity in the form of ready-made antirabies antibody to tide over the initial phase of the infection. Antirabies serum or RIG has the property of binding with the rabies virus, thereby resulting in the loss of infectivity of the virus.

Equine Rabies Immuno-globulin (ERIG):

ERIG is serum of heterologous origin raised by hyper-immunisation of horses. Hence these should be administered after sensitivity test. However, currently manufactured ERIG are highly purified and the occurrence of adverse events has been significantly reduced.

Human Rabies immunoglobulins (HRIG):

HRIG are free from the side effects encountered in a serum of heterologus origin, and because of their longer half life, are given in half the dose of equine antirabies serum. The antirabies sera should always be brought to room temperature (20 – 25°C) before use.

Dose of rabies immunoglobulin:

The dose of equine anti rabies serum is 40 i.u. per kg body weight of patient and is given after testing of sensitivity, up to a maximum of 3000 i.u. The ERIG produced in India contains 300 i.u. per ml. The dose of the human rabies immunoglobulins (HRIG) is 20 i.u. per kg body weight (maximum 1500 i.u.). HRIG does not require any prior sensitivity testing. HRIG preparation is available in concentration of 150 i.u. per ml.

Administration of immunoglobulin:

As much of the calculated dose of RIG as is anatomically feasible should be infiltrated into and around the wounds (Fig). Multiple needle injections into the wound should be avoided. Remaining, if any, after all wounds have been infiltrated, should be administered by deep intramuscular injection at an injection site distant from the vaccine injection site. Animal bite wounds inflicted can be severe and multiple, especially in small children. In such cases, the calculated dose of the rabies immunoglobulin may not be sufficient to infiltrate all wounds. In these circumstances, it is advisable to dilute the immunoglobulins in sterile normal saline 2 to 3 fold to be able to permit infiltration of all wounds. The total recommended dose of immunoglobulin must not be exceeded as it may reduce the efficacy of the vaccine

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If immunoglobulin was not administered when vaccination was begun, it can be administered up to the seventh day after the administration of the first dose of vaccine. Beyond the seventh day, Rabies Immunoglobulin (RIG) is not indicated since an antibody response to anti rabies vaccine is presumed to have occurred.

Immunoglobulin should never be administered in the same syringe or at the same anatomical site as vaccine.

Sensitivity test before administration of heterologous serum:

With anti-sera of equine origin, anaphylactic shock may occur and thus sensitivity testing is mandatory before giving ERIG. Skin test may be performed as per the manufacturers instructions given in the product insert. Otherwise general guidelines are described in Table 3.

Table 3 Skin testing prior to administration of ARS/ERIG

Inject 0.1 ml ARS/ERIG diluted 1:10 in physiological saline intradermally into the flexor surface of the forearm to raise a bleb of about 3 mm diameter.

Inject an equal amount of normal saline as a negative control on the flexor surface of the other forearm

After 15 minutes an increase in diameter to > 6 mm of wheal surrounded by flare is taken as positive skin test, provided the reaction on the saline test was negative

A negative skin test must never reassure the physician that no anaphylactic reaction will occur. Those administering ERIG should always be ready to treat early anaphylactic reactions with adrenalin. The dose is 0.5 ml of 0.1 percent solution (1 in 1000, 1mg/ml) for adults and 0.01 ml/kg body weight for children, injected subcutaneously or IM. If patient is sensitive to ERIG, HRIG should be used.

Approach to a patient requiring rabies immunoglobulin when none is available:

In circumstances where no immunoglobulin are available greater emphasis should be given to proper wound toileting followed by Essen schedule of Tissue culture vaccine with double dose on day 0 at 2 different sites intramuscularly (0 day – 2 doses on left and right deltoid, 3, 7, 14 and 28 days). It is emphasized that doubling the first dose of TCV is not a replacement to RIG.

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A full course of vaccine should follow thorough wound cleansing and passive immunization.

Tolerance and side effects:

With RIG, there may be transient tenderness at the injection site and a brief rise in body temperature which do not require any treatment. Skin reactions are extremely rare. *RIG must never be given intravenously since this could produce symptoms of shock, especially in patients with antibody deficiency syndromes.*

Serum sickness occurs in 1% to 6% of patients usually 7 to 10 days after injection of ERIG, but it has not been reported after treatment with HRIG.

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Active Immunization

Active immunization is achieved by administration of safe and potent TCVs. In India, NTV was used for post exposure treatment in public sector which was being given free of cost. However, as this vaccine was reactogenic, the production was stopped in December, 2004.

Intra-muscular Regimen (IM)

The currently available vaccines in India for IM administration are described in Table – 4.

Table 4: Anti Rabies Vaccines Currently available in India

Schedule

Name of the vaccine	Fixed virus strain	Substrate	Available
1. Cell Culture vaccines • Human Diploid Cell Vaccine (HDCV) • Purified Chick Embryo Cell Vaccine (PCEC) • Purified Vero Cell Rabies Vaccine (PVRV)	• Pitman Moore (PM) • LEP-Flury • Pitman Moore (PM)	• MRC-5 • Primary SPF chick embryo cells • Vero Cells	• Produced locally in private. Sector • Produced locally in private sector • Imported + produced locally in public & private sector
2. Purified Duck Embryo Vaccine	Pitman Moore (PM)	Duck Embryo	Produced locally in private Sector

Essen Schedule:

Five dose intramuscular regimen - The course for post exposure prophylaxis should consist of intramuscular **administration of five injections on days 0, 3, 7, 14 and 28**. The sixth injection (D90) should be considered as optional and should be given to those individuals who are immunologically deficient, are at the extremes of age and on steroid therapy. Day 0 indicates date of first injection. This schedule is followed in India.

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Zagreb schedule:

Abbreviated multisite intramuscular regimen (2-1-1) – One dose of vaccine administered intramuscularly in the left and one into right deltoid region on day 0 followed by one dose on days 7 and 21 in deltoid region. This schedule saves 2 clinic visits and one vaccine dose.

Indications.

All age groups of animal bite victims of Category II and III require the same number of injections and dose per injection. The Category III exposures, in addition require administration of rabies immunoglobulins as discussed earlier.

Site of inoculation.

The deltoid region is ideal for the inoculation of these vaccines. Gluteal region is not recommended because the fat present in this region retards the absorption of antigen and hence impairs the generation of optimal immune response.

Storage and transportation.

Though most tissue culture vaccines are marketed in freeze dried (lyophilized) form which is more tolerant of vagaries of temperature, yet it is recommended that these vaccines should be kept and transported at a temperature range of 2-8°C. Freezing does not damage the lyophilized vaccine but there are chances of breakage of ampoule containing the diluent. Liquid vaccines should never be frozen.

Reconstitution and storage.

The lyophilized vaccine should be reconstituted with the diluent provided with the vaccine immediately prior to use. *However, in case of unforeseen delay it should not be used after 6-8 hours of reconstitution.*

Adverse effects with tissue culture vaccines.

The tissue culture vaccines are widely accepted as the least reactogenic rabies vaccines available today. Various studies have now shown that adverse effects can be either general in nature or allergic in origin. The general adverse reactions include sore arm, headache, malaise, nausea, fever and localised Oedema at the site of injection. Symptomatic treatment may be needed.

Switch over from one brand of vaccine to the other:

Shifting from one brand of TCV to other brand should not be encouraged as literature supports that good immunity is best achieved with same brand.

Intra-dermal (ID) Regimens

Concept of Intra-dermal inoculation of Anti rabies Vaccines (IDRV)

Intra-dermal regimens consist of the intra-dermal administration of a fraction of intramuscular dose of certain rabies vaccine on multiple sites. The vaccines used are same; however route, dose and site of administration differs. The use of intra-dermal route leads to considerable savings in terms of total amount of vaccine needed for full pre- or post- exposure vaccination, thereby reducing the cost of active immunization.

Intra-muscular route (IM):

Single bolus dose (1ml) of Rabies vaccines/antigen when given by IM route gets deposited in the muscles. There after the antigen is absorbed by the blood vessels and is presented to antigen presenting cells which triggers immune response.

Intra-dermal route (ID):

Small amount (0.1ml) of Rabies vaccines/antigen is deposited in the layers of the skin at multiple sites. The antigen is directly presented to the antigen presenting cells (with out circulation/dilution in blood) at multiple sites triggering a stronger immune response.

Mechanism of Action of IDRV

Intra-dermal inoculation is deposition of approved rabies vaccine (or antigen) in the layers of dermis of skin. Subsequently the antigen is carried by antigen presenting cells via the lymphatic drainage to the regional lymph nodes and later to the reticulo-endothelial system eliciting a prompt and highly protective antibody response. Immunity is believed to depend mainly upon the CD 4 + T- cell dependent neutralizing antibody response to the G protein. In addition, cell-mediated immunity has long been reported as an important part of the defense against rabies. Cells presenting the fragments of G protein are the targets of cytotoxic T- cells and the N protein induced T helper cells. The immune response induced by IDRV is adequate and protective against rabies.

International Scenario

Multiple clinical trials since early eighties proved that ID route of inoculation of Tissue Culture Anti-Rabies Vaccines (TCARV) is efficacious, safe and economical. It requires less quantity of vaccine which brings down the cost of immunization and allows wider coverage in the existing quantity of vaccine. WHO recommended use of ID route of inoculation of TCARV in 1992.

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Thailand, Sri Lanka and Philippines have successfully adopted ID route of inoculation.

Schedules approved by WHO for ID route

1) Updated Thai Red Cross Schedule (2-2-2-0-2):

Vaccines approved for TRC schedule:

- Purified Vero cell Rabies Vaccine (PVRV) produced by Aventis Pasteur
- Purified Chick Embryo Cell rabies Vaccine (PCECV) produced by Chiron Vaccines

Regimen: 2-2-2-0-2 i.e. one dose of vaccine, in a volume of 0.1ml is given intradermally at two different lymphatic drainage sites, usually the left and right upper arm, on days 0, 3, 7 and 28. (Used in Guideline)

2) Eight site intradermal regimen (8-0-4-0-1-1)

Vaccines approved for eight site schedule:

- Human Diploid Cell Vaccine (HDCV) produced by Aventis Pasteur

Purified Chick Embryo Cell rabies Vaccine (PCECV) produced by Chiron Vaccines **Regimen:** 8-0-4-0-1-1 i.e. one dose of 0.1 ml is administered intradermally at eight different sites (Fig.-5) (upper arms, lateral thighs, suprascapular region and lower quadrant of abdomen) on day 0. On day 7, four 0.1ml injections are administered intra-dermally into each upper arm (deltoid region) and each lateral thigh. Following these injections one additional 0.1ml dose is administered on 28 and 90

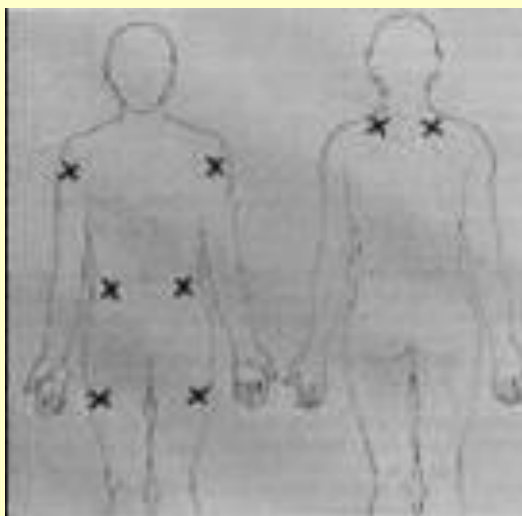


Fig 5. Sites for id inoculation in 8-site id regimen

General guidelines

- Vaccines to be applied by intradermal route of administration should meet the WHO requirements for production and control related to vaccines for intramuscular use including an NIH potency test of at least 2.5IU per single dose (intramuscular).
- Immunogenicity and safety of vaccine in question should be demonstrated in appropriate human trials using WHO post-exposure regimens.
- In countries where national health authorities have approved the ID route for pre-post exposure prophylaxis, and for vaccines that can be used by that route, the vaccine package leaflet should include a statement indicating that the potency as well as immunogenicity and safety allow safe use of vaccine by ID pre- and post-exposure prophylaxis in addition to other relevant information as described in WHO requirements for vaccine production and control.
- The decision to implement economical ID post-exposure prophylaxis rests with government agencies that define rabies prevention and treatment policies in their own countries.
- Intradermal injections must be administered by staff trained in this technique.
- Vaccine vials must be stored at 2^o to 8^oC after reconstitution
- The total content of reconstitute vial should be used as soon as possible, but at least within 8 hours.
- All the reconstitute vaccines should be discarded after 8 hours of reconstitution and at the end of the day
- Rabies vaccines formulated with an adjuvant should not be administered intradermally.
- Vaccine when given intra-dermally should raise a visible and palpable bleb in the skin.
- In the event that the dose is inadvertently given subcutaneously or intramuscularly, a new dose should be given intradermally.

IDRV Vaccines and regimens approved for use in the country

The following vaccines have been approved by National Authorities in first phase for use by intradermal route after assessing the safety, efficacy and feasibility data as recommended by WHO:

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Vaccines

PVRV – Verorab, Aventis Pasteur (Sanofi Pasteur) India Pvt. Ltd.

PCEC – Rabipur, Chiron Behring Vaccines Pvt. Ltd.

PVRV – Pasteur Institute of India, Coonoor

PVRV – Abhayrab, Human Biologicals Institute.

Potency of approved vaccines:

The vaccines should have stated potency of ≥ 2.5 IU per IM dose, irrespective of reconstituted volume. The same vaccine is used for ID administration as per stated schedule. 0.1ml of vaccine, irrespective of reconstituted volume, is administered per id site as per schedule below.

Pre-exposure prophylaxis: This involves injection of 0.1 ml of reconstituted vaccine per id site on days 0, 7 and 21 or 28.

Maintenance of vaccine vial in use

- Use aseptic technique to with draw the dose
- Store in a refrigerator at 2°C to 8°C
- Reconstituted vaccines should be used as soon as possible or within 6 to 8 hours if kept at 2°C to 8°C. All unused reconstituted vaccine at the end of 6-8 hours must be discarded.

Materials required

A vial of rabies vaccine approved for IDRV and its diluent.

2 ml disposable syringe with 24 G needle for reconstitution of vaccine.

Disposable 1 ml (insulin) syringe (with graduations up to 100 or 40 units) with a fixed (28 G) needle (Fig.6)

Disinfectant swabs (e.g. 70% ethanol, isopropyl alcohol) for cleaning the top of the vial and the patients' skin.



Fig.6 : 1ml syringe with hypodermic needle (Insulin syringe)

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ID injection technique

- Using aseptic technique, reconstitute the vial of freeze-dried vaccine with the diluent supplied by the manufacturer.
- With 1 ml syringe draw 0.2 ml (up to 20 units if a 100 units syringe is used or up to 8 units if a 40 units syringe is used) of vaccine needed for one patient (i.e. 0.1 ml per ID site X 2 sites 0.2 ml)
- Expel the air bubbles carefully from the syringe thereby removing any dead space in the syringe.
- Using the technique of BCG inoculation, stretch the surface of the skin and insert the tip of the needle with bevel upwards, almost parallel to the skin surface (Fig.7) and slowly inject half the volume of vaccine in the syringe (i.e. 0.1ml; either 10 or 4 units) into the uppermost dermal layer of skin, over the deltoid area, preferably an inch above the insertion of deltoid muscle.
- If the needle is correctly placed inside the dermis, considerable resistance is felt while injecting the vaccine.
- A raised papule should begin to appear immediately causing a peau d' orange (orange peel) appearance (Fig.8). Inject the remaining half the volume of vaccine (i.e. 0.1ml; either 10 or 4 units) on the opposite deltoid area.
- If the vaccine is injected too deeply into the skin (subcutaneous), papule is not seen.
- Then the needle should be withdrawn and reinserted at an adjacent site and the ID vaccine given once more.



Fig.7 Insertion of needle for ID
inoculation



Fig. 8 Bleb raised on ID inoculation

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Anti-rabies treatment centers which meet the following criteria may use ID administration:

- Have adequately trained staff to give id inoculation of antirabies vaccine
- Can maintain cold chain for vaccine storage
- Ensure adequate supply of suitable syringes and needles for id administration
- Are adequately well versed in management of open vial and safe storage practices.

General Considerations for Implementation

- The vaccine which is being used for IM administration can be used for ID administration after the vaccine manufacturers amend the label and package insert to indicate that the vaccine is fit for IM and ID route in designated centers. The manufactures should get the same approved by DCGI.
- A copy of protocol to generate Post Marketing Surveillance (PMS) data should be prepared and submitted by vaccine manufacturers for approval from DCGI's office. PMS should be maintained for minimum of two years.
- Since inclusion in Indian Pharmacopeia takes a long time, with the approval of vaccine from DCGI office and undertaking from the Pharmaceutical firms that the vaccine is fit for use by IM and ID route, the State health authorities may start use of ID route of inoculation of anti-rabies vaccines.

Considering the recommendations on intradermal application of rabies by WHO and results of safety, efficacy and feasibility trials conducted in India, Drug Controller General of India (DCGI) approved the use of reduced dosage intradermal vaccination regimen for rabies post-exposure prophylaxis. The use of this route leads to considerable savings in terms of the total amount of vaccine needed for a full post-exposure vaccination, thereby reducing the cost of active immunization.

Precautions while using ID regimen:

When the intradermal route is used, precautions include staff training, conditions and duration of vaccine storage after reconstitution, use of appropriate 1 ml syringe and short hypodermic needles. Vaccines to be applied by intradermal route of administration should meet WHO requirements for production and control related to vaccines for intramuscular use, including an NIH test potency of at least 2.5 IU per single (intramuscular) dose. In addition, immunogenicity and safety of the vaccine in

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question should be demonstrated in appropriate human trials using WHO/national post-exposure prophylaxis regimens. The vaccine package leaflet should include a statement indicating that the potency as well as immunogenicity and safety allow safe use of the vaccine for intradermal post-exposure prophylaxis, in addition to other relevant information as described in the WHO requirements for vaccine production and control.

Post exposure therapy for previously vaccinated persons

Managing re-exposure following post-exposure treatment with TCV:

If re-exposed, persons who have previously received full post-exposure treatment (either by IM or ID route) with a potent cell-culture vaccine should be given only two booster doses, intramuscularly/intra-dermally (0.1 ml at 2 sites) on days 0 and 3, but no rabies immunoglobulin.

Managing exposure following pre-exposure prophylaxis with TCV:

If after recommended pre-exposure prophylaxis, a vaccinated person is exposed to rabies, a proper wound toileting should be done and two IM/ID (0.1 ml at 2 sites) doses of Tissue Culture Vaccine be given on days 0 and 3. Treatment with RIG is not necessary

Managing re-exposure following post exposure treatment with NTV:

Persons who have previously received full post-exposure treatment with NTV should be treated as fresh case and may be given treatment as per merits of the case.

Protective level of anti-rabies antibody.

Human antibodies are believed to play important role in protection against rabies and a titer of 0.5 i.u./ml or more in serum is considered as protective.

Role of laboratory facilities in management of animal bite cases

In countries like India where rabies is endemic, the animal bite management should not depend on laboratory results. The treatment should be started immediately as per recommended guidelines.

Pre-exposure prophylaxis

Pre-exposure prophylaxis may be offered to high risk group like laboratory staff handling the virus and infected material, clinicians and para-medicals attending to hydrophobia cases, veterinarians, animal handlers and catchers, wildlife wardens, quarantine officers and travelers from rabies free areas to rabies endemic areas. Pre-exposure vaccination is administered as one full dose of vaccine intramuscularly or 0.1 ml intradermally on days 0, 7 and either day 21 or 28.

Laboratory staff and others at high continuing risk of exposure should have their neutralizing antibody titers checked every 6 months. If it is less than 0.5 i.u./ml a booster dose of vaccine should be given. Such individuals on getting exposed to rabies virus after successful pre-exposure immunization require only two booster injections of vaccine given on days 0 and 3 without any anti rabies serum.

National rabies Control Program

Government of India launched a pilot project on prevention and control of Rabies in 5 selected cities in 11th Five Year Plan. With the lessons learnt from the pilot project, Government of India launched Under XII plan, National Rabies Control Programme (NRCP) in 12th Five Year Plan with objectives of

- To reduce human deaths due to rabies.
- To implement a consensus strategy for control of rabies in dogs to cut down the transmission of rabies

As control of rabies involves two components viz. elimination of human rabies deaths and control of canine rabies to break down the transmission, national rabies control program has two components

- Human component
- Animal component

Strategies adopted for the human component are

- Training of health professionals
- Implementing use of intra-dermal route of inoculation of cell culture vaccines
- Strengthening surveillance
- IEC
- Laboratory strengthening of four regional laboratories

The strategies adopted for Animal Component are

- Population survey of dogs
- Mass vaccination of dogs
- Dog population management
- Strengthening surveillance and response

Nodal Agency -

Human Component: Nodal agency is National Centre for Disease Control (NCDC), Director GHS, MoHFW. The human component will be implemented in all the States and UTs. Ministry of Health NAD FAMILY Welfare is providing grant- in aid under Memorandum of Understanding to the states / UTs governments for carrying out the activities of human component

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Animal Component: Nodal agency is Animal Welfare Board of India (AWBI), Min of Environment & Forest – is being pilot tested in Haryana & Chennai. Ministry of Health & Family Welfare is providing the funds under Memorandum of Understanding to AWBI to carry out various activities

Roles and responsibilities of State Nodal Officers:

State Nodal Officer will be officer in-charge to implement the activities of human component of National Rabies Control Program in their respective states The roles and responsibilities of State Nodal Officers are as under:

1. To coordinate with State Health Officials for implementation of activities human component of National Rabies Control Program
2. To ensure utilization of funds as per the guidelines and submit regular Utilization Certificate to the nodal center.
3. To strengthen surveillance of animal bite management and rabies and transmission of monthly report to the nodal centers as per the format provided
4. To organize training of the different categories of the health staff on animal bite management as per the National Guidelines on Rabies Prophylaxis
5. To monitor and evaluate the program activities and collection of base line data as per format provided including mapping of health institutes/center with facilities for treatment of Animal bite victims and rabies cases and transmission of the same to the nodal center.
6. To carry out the IEC / BCC activities on rabies and animal bite management awareness
7. To provide regular feedback about the program activities to the nodal center.
8. To ensure regular supply of the DGCI approved
9. Anti-Rabies Vaccines for ID inoculation and ERIG at all ARCs in their states
10. To identify laboratories for strengthening rabies diagnostics.
11. To liaise with State Government and State Animal Husbandry department regarding status of activities being undertaken for prevention and control of rabies in animals.

Operational Guidelines:

The operational guidelines for implementation of the activities of human component of National Rabies Control Program are as under:

Identification of District Focal Point

All the State Nodal officers are advised to coordinate with the concerned District Health officials to identify a District Focal Point (may be District Surveillance

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Officers or district epidemiologist), responsible for district level coordination of activities of human component of the program. The District focal point will be responsible for collection/ compilation and collation of data of all Anti Rabies Clinics in the district (PHC/ CHC/ District level/ tertiary level health facilities). Data from all the district of the state may be compiled at the state level and summary report may be forwarded to the NCDC, Zoonosis Division.

Capacity building:

Trainings of the Health Professionals on appropriate animal bite management should be carried out as per national guidelines on rabies prophylaxis

Trainees : Each state is envisaged to train different categories of the health staff (medical officers, Nurses, pharmacist etc) of PHC, CHC, District hospitals, state level tertiary care institutes and all designated Anti Rabies Clinic officials , where the facilities for treating the Animal Bite Victims are available.

The trainings would be facilitated by the master trainers trained at the NCDC. The NCDC officials may also be invited for lectures.

The trainings should focus on appropriate animal bite management and ID route of inoculation for post-exposure prophylaxis. The training curriculum is provided at Annex-II. The power point presentations may be prepared using the new revised National Guidelines for Rabies Prophylaxis , which are freely downloadable from NCDC website ([www//http: ncdc.gov.in](http://www.ncdc.gov.in))

The trainings may be organized at identified training institutes/centers of the states.

Implementation of Intra Dermal (ID) route of inoculation for post-exposure prophylaxis:

Implementation of ID route of inoculation for rabies post exposure prophylaxes is one of the important objective of the National Rabies Control Program. The use of intra-dermal route leads to considerable savings in total amount of vaccine needed for full post- exposure vaccination, thereby reducing the cost of active immunization. All the state nodal officers should ensure the practice of ID route of inoculation for Anti Rabies vaccine in all major Anti Rabies Clinics of the State. The SNO should ensure the regular availability of DCGI approved anti-rabies vaccines for ID Route of inoculation and Equine Rabies Immunoglobulin (ERIG)

Strengthening of rabies diagnostic facilities:

State Nodal Officers are advised to map the labs where the rabies diagnostic facilities are available. A brief report of the current status of rabies diagnostic

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facilities in the existing labs of Govt. medical colleges/ tertiary care health facilities of the states should be prepared.

Anti-Rabies Clinics (ARC)

Anti-Rabies Clinics/Centers are the health facilities manned by trained doctor/s and nurse/s where individuals with rabies exposure are evaluated and managed. Strengthening of the existing Anti rabies Clinic and establishment of Newer ARC based on community need assessment is an important objective of the national rabies control program.

State Nodal Officers are advised to do mapping of existing district wise ARC in the Government Health facilities (PHC/ CHC/ District Hospital or other Tertiary level health institute) as well as list of all private health facilities where there is facility of treatment of animal bite victims or indoor treatment facility for rabies cases. The need assessment for establishing newer ARC at the identified health facilities may simultaneously be carried out by SNO. The Model ARC should meet the following minimum requirements -

Facilities at ARC:

Minimum facilities envisaged at ARC are as under:-

- Management of Animal Bite Wounds- wound washing facility
- Availability of Rabies post exposure prophylaxis- Anti-rabies vaccine and ERIG
- Functional referral services for hydrophobia cases
- Standardized recording and reporting systems

Physical Infrastructure: the Physical infrastructure of the ARC should meet the following minimum requirements

- Visible signage at the entrance as well as outside the center
- Visible organizational Chart
- Time schedule (functional hrs. of ARC)
- Visible flow chart/ algorithm of “ decision to treat” (available at pg -28 of national guidelines of rabies prophylaxis Decision Tree: Guide to Post-Exposure Prophylaxis)
- Visible IEC messages
- Separate Wound washing facility with preferably continuous tap water (the water available should be safe and clean). If the tap water is not available the water should be stored in clean covered bucket)
- Refrigerator with a calibrated thermometer, exclusive for vaccine/ RIG storage
- Vaccine carrier for temporary storage

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- Facility for proper Biomedical waste management with availability of Color coded waste bins and sharp boxes
- Weighing scale

Logistics:-

1. Equine Rabies Immunoglobulin
2. Tissue culture anti-rabies vaccine approved by DCGI for ID/IM route.
3. Consumable : self-mounted insulin syringes (AD), Dressing Kits, Soap and Gloves
4. IV Fluids and Emergency drugs for adverse reaction

Recording and reporting formats: Following recordings and reporting formats should exist at ARC -

1. Animal bite exposure register (Annex -III)
2. Rabies PEP CARD (in duplicate one for the Victim and one for the ARC record (Annex -IV)
3. ARC monthly monitoring formats, NRCP –M03 (Annex - V)
4. Copy of “National Guidelines for Rabies Prophylaxis”, available at NCDC website [www//http//ncdc.gov.in](http://ncdc.gov.in))
5. SOP for sample collection for anti-rabies antibody titer estimation ((Annex-VI)

Information Education and Communication:

Rabies is a disease of both rural and urban areas. However, most rabies patients do not seek proper PEP in health-care facilities, and many visit traditional healers. Community participation is an essential element of any disease control programme. It is envisaged that communities should be made aware of do's and don'ts of animal bite management and proper washing of wounds after animal bite. The prototype IEC material has been developed following expert consultation by NCDC which may be used for the IEC activities in the state after necessary modification according to local need. Prototype IEC Material is provided at Annexure - VIII

Human Rabies Information System:

The existing CBHI portal captures the data on human deaths due to rabies from across the country. The weekly dog bite data is also available on the IDSP portal. It is envisaged under NRCP to further strengthen the reporting of dog bites and rabies deaths at IDSP and CBHI portal respectively.

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To further strengthen the recording and reporting of animal bites and rabies cases, all the States will be generating regular reports at each ARC located at PHC, district level or tertiary care health facility.

Record maintenance at Anti Rabies Clinic:-

Each Anti Rabies Clinic will maintain a minimum set of information on each patient, as reflected in the attached individual Rabies PEP Treatment Record (Annexure -IV), that shall be readily available for monitoring and evaluation. A Rabies PEP Treatment Card, as prescribed by the MO I/C of ARC will contain the patient details, history of exposure and record of immunizations being provided. The PEP card will be issued to each patient and the duplicate of the same will be maintained at the respective ARC. Each ARC will maintain an Animal bite exposure register (Annexure-III) and will generate the monthly report as per the NRCP-M03 format (Annexure -V) and will submit the same to the District Focal Point within 5 days of the following month.

The following tables summarizes the reports to be generated and flow of information for management of the activities of human component of National Rabies Control Program.

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- A. Base line information on status of Anti-rabies Centers(ARC) and management of animal bite victims-** to assess the present status of Anti Rabies Clinic and facilities for Animal Bite Management following base line information is required to be collected at the District and State level

DATA	LEVEL	FORMATS	FROM - TO	PERIODICITY
District base line information	District level	NRCP-BLD (Annex - IX)	District Focal Point to State Nodal Officer	At the beginning of the program
State base line information	State level	NRCP –BLS (Annex- X)	State Nodal Officer to Nodal Agency (NCDC)	At the beginning of the program

B. Monthly information on animal bite victims and rabies vaccination

DATA	LEVEL	FORMATS	FROM - TO	PERIODICITY
ARC monthly report	Anti-Rabies Clinics	NRCP-M03 (Annex - V)	ARC MO I/C to District Focal Point	Monthly (within 5 days of the following month)
District monthly report	District	NRCP-M02 (Annex - XI)	District Focal Point to State Nodal Officer	Monthly (within 10 days of the following month)
State monthly report	State	NRCP-M01 (Annex - XII)	State Nodal Officer to Nodal Agency (NCDC)	Monthly (within 15 days of the following month)
Hydrophobia/ rabies monthly report	State	NRCP –RC (Annex - XIII)	State Nodal Officer to Nodal Agency (NCDC)	Monthly (within 15 days of the following month)

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ANNEXURE -V

NATIONAL RABIES CONTROL PROGRAM

ANTI RABIES CLINIC MONTHLY REPORT (NRCP-M03)

**Monthly report of Animal Bite Victims receiving treatment at Anti Rabies Clinic
(to be submitted by ARC MO I/C to District Focal Point)**

District Focal point name & complete address:

Name of ARC :

Address:

Month & Year of reporting:

Total no. of patients attending ARC		
Biting animal		
Dog		
Cat		
Monkey		
Any other (specify)		
Category of bite		
I. Licks on intact skin		
II. Minor scratches or abrasions without bleeding		
III Single or multiple bite with bleeding or Contamination of mucous membrane		
Any other (specify)		
Wound washing facility at ARC	Yes/No	
Vaccine used at ARC(monthly)		
Quantity received + Stock of previous month		
Quantity utilized		
Mention shortage if any		
ARS used at ARC(monthly)		
Quantity received + Stock of previous month		

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Quantity utilized		
Mention shortage if any		
Vaccination given by		
IM route (Essen schedule on day 0,3,7,14,28)		
ID route (update Thai Red Cross Regimen : 2-2-2-0-2)		
Category III victims given ARS		
Any other remarks		
Signature:		
Date:		

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ANNEXURE -IX				
<i>National Rabies Control Program</i>				
Baseline Data Collection at District Level (NRCP –BLD)				
State name:	District name:			
Name of SNO:	Name of District Focal Point			
Address	Address			
Contact no	Contact no			
Email ID	Email ID			
District Population: TotalUrban.....Rural				
Total No of ARC in the District :				
Details of ARC in the District	ARC - 1	ARC- 2	ARC- 3	ARC- 4
Name and Address of the Hospital/ Health Facility where ARC is located (Level of HF whether located at Tertiary care/ CHC/ PHC/ ESI/ Railways				
Functional Hrs. of ARC (FromAM to PM)				
Staff Position at ARC (fully Dedicated)				
Doctor				
Paramedical staff				
Training of staff in Animal Bite Management and ID Route of inoculation				
Doctor				
Paramedical staff				

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Infrastructure / basic facilities at ARC				
Separate Wound washing facility (Y/N)				
Availability of running tap water (Y/N)				
Area designated for management of wound (Y/N)				
Area designated for administering PEP to bite victim (Y/N)				
Visible signage (Y/N)				
Visible Flow Charts/ algorithm (Y/N)				
Display of IEC material (Y/N)				
Availability of ARV Y/N				
Availability of ARS Y/N				
State reason if any shortage				
AD syringes Y/N				
Antiseptics/ Antibiotics Y/N				
IV set / IVF				
Emergency Drugs to treat Adverse Drug Reactions Y/N				
Dressing material Y/N				
Waste disposal facility (Colored bins) Y/N				
Data of Animal Bite Victims at ARC during last three years				
Attendance at ARC				
(1st Jan to 31st Dec Each Year)				
No of Dog Bite cases				
No of other Animal Bite cases				
No of Cat -III bites				
No of Cat -II bites				
No of Cat -I bites				
No of cases given PEP by IM route				
No of cases given PEP by ID route				

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No of Cat III bites given RIG				
Availability of ARV and ERIG in the District : provide details of				
Vaccine supply				
Name of the vaccine				
How it is supplied				
Quantity supplied (yearly)				
Shortage if any				
Anti-Rabies serum Supply				
Regular/ irregular supply				
If regular Quantity supplied (yearly)				
Shortage if any				
List of ID/Tertiary care hospital in the District (hospitals admitting suspected rabies cases):-				
Is there a need to establish newer ARC in the district? If so, give details.				

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ANNEXURE-XI			
NATIONAL RABIES CONTROL PROGRAM			
District Monthly Report (NRCP-M02)			
Compiled Monthly Report of Animal Bite Victims receiving treatment at Anti Rabies Clinic (to be submitted by District Focal Point to State Nodal Officer)			
State name :			
Name of the District			
Total no. of patients attending ARC	ARC 1	ARC 2	ARC 3
Biting animal			
Dog			
Cat			
Monkey			
Any other (specify)			
Category of bite			
I. Licks on intact skin			
II. Minor scratches or abrasions without bleeding			
III Single or multiple bite with bleeding or Contamination of mucous membrane			
Any other (specify)			
Wound washing facility at ARC			

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Vaccine used at ARC(monthly)			
Quantity received + Previous Stock			
Quantity utilized			
Mention shortage if any			
ARS used at ARC			
Quantity received + Previous Stock			
Quantity utilized			
Mention shortage if any			
Vaccination given by			
IM route (Essen schedule on day 0,3,7,14,28)			
ID route (update Thai Red Cross Regimen : 2-2-2-0-2)			
Category III bites given ARS			
Any other remarks			
Signature :			
Date:			

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ANNEXURE -IV		
<i>NATIONAL RABIES CONTROL PROGRAM</i>		
<i>RABIES POST EXPOSURE TREATMENT CARD (To be retained at ARC)</i>		
Name and address of the health facility :		
Reg. No		
Name	Patient Address	
Age/ Sex	Contact No	
Weight (kg)	Date treatment started (MM/DD/YYYY)	
Category of Exposure	Past h/o of animal bite	
I. Licks on intact skin	If Yes specify date	
II. Minor scratches or abrasions without bleeding		
III Single or multiple bite with bleeding or Contamination of mucous membrane	Past h/o vaccination	
	If Yes specify date	

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Date of Exposure/bite		Specify whether Partial/ complete () Yes () No	
Site of Bite/ Bites			
Type of animal		Biting animal status	
Dog	Monkey	Alive	
Cat	Others	Dead	
Wound management			
Washed immediately (by patient) () Yes () No		Wound washed at facility () Yes () No	
Antiseptic application () Yes () No		ERIG Infiltration () Yes () No	
Post exposure vaccination record Route of Administration () ID () IM			
Period	Date due	Date given	signature
Day 0			
Day 3			
Day 7			
Day 14			
Day 28			
<u>RABIES POST EXPOSURE TREATMENT CARD (VICTIM COPY)</u>			
Name and address of the health facility :			
Reg. No			
Name		Patient Address	
Age/ Sex		Contact No	

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Weight (kg)	Date treatment started (MM/DD/YYYY)	
Category of Exposure	Past h/o of animal bite	
I. Licks on intact skin	If Yes specify date	
II. Minor scratches or abrasions without bleeding		
III. Single or multiple bite with bleeding or Contamination of mucous membrane	Past h/o vaccination	
	If Yes specify date	
Date of Exposure/bite	Specify whether Partial/ complete () Yes () No	
Site of Bite/ Bites		
Type of animal	Biting animal status	
Dog Monkey	Alive	
Cat Others	Dead	
Wound management		
Washed immediately (by patient) () Yes () No	Wound washed at facility () Yes () No	
Antiseptic application () Yes () No	ERIG Infiltration () Yes () No	
Post exposure vaccination record Route of Administration () ID () IM		
Period	Date due	Date given
Day 0		
Day 3		

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Day 7				
Day 14				
Day 28				

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ANNEXURE-III

NATIONAL RABIES CONTROL PROGRAM

ANIMAL BITE EXPOSURE REGISTER

REGISTRATION		HISTORY OF EXPOSURE								POST EXPOSURE PROPHYLAXIS					Biting animal status dead/alive after 10 days	Remarks
No	Date	Name of the Patient	Address	Age (yrs)	SEX	Date of bite	Site of bite	Biting animal species	Site	Category of bite	Washing Y/N	ERIG Y/N	Route ID/IM	Brand name		

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Epidemic Branch
Commissionerate of Health, Medical Services and Medical Education (H.S.)
Gandhinagar